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When the Dashboard Knows More Than the Report: Using AI-Powered Business Intelligence to Surface Hidden Data Errors in Financial Services

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***Index Terms*—**AI-assisted financial data analysis, Amazon QuickSight, data reconciliation, algorithm aversion, fintech project management, technology adoption.

Abstract

This article describes the deployment of Amazon QuickSight—Amazon Web Services' AI-powered business intelligence platform—to analyze transaction and position data reconciliation reports in a financial services environment. In a dataset of approximately 84,000 transactions over a six-week reporting cycle, the tool flagged 203 raw discrepancies compared with 147 from the manual process, a 38% increase in raw detection. Following analyst-led reclassification of timing artifacts, 172 errors were validated, representing a net increase of approximately 17% over the manual baseline. Root-cause analysis further revealed that 71% of validated errors originated from two upstream system integration points that the manual process had never isolated. The investigation that followed, and the organizational resistance it encountered, reflects patterns well-documented in algorithm aversion research. Experienced analysts, whose professional identity was tied to manual reconciliation expertise, initially rejected the tool's findings—consistent with Dietvorst, Simmons, and Massey's demonstration that people lose confidence in algorithms more rapidly than in human forecasters after observing the same error. This article reports what happened, what was learned, and translates those lessons into practical guidance for engineering and technology managers across organizations of varying size: large financial institutions, small- and medium-sized enterprises, and fintech startups. The central argument is that AI-assisted reconciliation tools change not only the speed of reporting but the nature of insight that becomes available—and capturing that change requires deliberate management action, not just technology deployment.

I. Introduction and Prior Work

Data reconciliation—the systematic comparison of transaction records, position data, and account balances across systems—sits at the operational heart of financial services. Regulatory bodies including the SEC (Rule 17a-3 and 17a-4) and the FDIC impose specific requirements on the accuracy and timeliness of reconciliation reporting for institutions handling customer deposits, securities positions, and investment accounts [1], [2]. These requirements exist because reconciliation errors carry direct downstream consequences: misstated client positions, inaccurate filings, and in severe cases, regulatory action and mandatory disclosure obligations.

Despite this regulatory weight, reconciliation in many organizations remains predominantly manual. A 2021 industry survey found that finance teams spend up to 50% of their working time on

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data reconciliation and validation rather than higher-value analysis [3]. Manual processes are not only slow; they carry structural blind spots. Analysts reviewing line-item exports are constrained by the volume of data they can examine within a reporting cycle. Errors that appear only when patterns are examined across the full dataset—rather than within individual reports—frequently go undetected.

AI-powered business intelligence platforms such as Amazon QuickSight, Microsoft Power BI, and Tableau offer pattern recognition at a scale that manual review cannot replicate. Yet technology adoption research consistently shows that introducing such tools into expert workflows generates resistance that is neither irrational nor surprising. Dietvorst, Simmons, and Massey [4] demonstrated that people lose confidence in an algorithm more rapidly than in a human after observing both make the same mistake—a phenomenon they termed "algorithm aversion." Critically, this effect persisted even when the algorithm outperformed the human forecaster. In parallel, Logg, Minson, and Moore [5] showed that algorithm aversion is most pronounced among domain experts, whose professional identity is tied to the judgments the tool appears to challenge. Both findings map directly onto the experience described in this article.

Leonardi [6] adds a complementary organizational lens: misalignments between how employees interpret a new technology through social interaction and how they actually experience it through use generate negative assessments that impede adoption, even among those who endorse the tool's broader goals. Together, these three bodies of work explain why introducing an AI reconciliation tool into an expert analyst team is a predictably contested event—and why managing that contest is as important as configuring the tool correctly.

This article contributes a practitioner account of how these dynamics played out in a real deployment and derives from that account a structured set of recommendations for managers. The gap addressed is specific: there is limited published practitioner guidance on how engineering and technology managers should prepare their teams, interpret AI-generated findings against regulatory obligations, and structure governance when moving from manual to AI-assisted reconciliation in regulated financial environments. The article also responds to Brem's [8] call for EMR contributions that translate practitioner experience into immediately actionable guidance for working managers.

II. Methodology

The insights in this article are drawn from the author's direct experience as a Senior Data Manager responsible for transaction and position data reconciliation at a mid-sized registered investment advisory firm. The deployment described involved implementing Amazon QuickSight as an AI-assisted supplement to—and eventual replacement for—a long-standing manual reconciliation workflow covering approximately 84,000 transactions per six-week reporting cycle.

Observations were gathered over approximately six months spanning pre-deployment baseline measurement, initial rollout, recalibration following resistance, and stabilized operation. Error counts were drawn from system-generated reconciliation logs; analyst reclassifications were documented in a shared tracking sheet maintained throughout the deployment. Time estimates reflect logged hours from the team's project management tool and should be understood as directional rather than formally audited figures. The author was both observer and participant in this deployment, which introduces the possibility of confirmation bias—a limitation acknowledged explicitly here.

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This is a single-organization practitioner case. It does not claim statistical generalizability. Readers should evaluate the SME and startup guidance in Section V as reasoned extrapolation from the case context and the regulatory landscape, not as case-derived findings, and should calibrate accordingly.

III. The Deployment: What QuickSight Found

The manual reconciliation process produced error reports by having analysts review transaction and position system exports, compare them against expected values, and flag discrepancies for follow-up across a six-week reporting cycle. In the baseline period, this process identified 147 discrepancies per cycle across approximately 84,000 transactions.

When QuickSight was connected to the same dataset, it flagged 203 raw discrepancies in the first full six-week reporting cycle—a 38% increase over the manual count. Analyst review subsequently reclassified approximately 15% of those flags (roughly 31 items) as timing artifacts: expected differences in when transactions settle across systems rather than genuine errors. The validated error count following reclassification was therefore approximately 172, representing a net increase of roughly 17% over the manual baseline.

An important caveat applies to this comparison: the manual process was not designed to distinguish timing artifacts from genuine errors, meaning the baseline figure of 147 may itself have included some unidentified timing artifacts. The two figures are therefore not strictly like-for-like. The 17% net increase in validated errors is the more defensible headline; the 38% raw figure reflects the difference in detection capacity before human validation is applied.

QuickSight's drill-down analysis further revealed that 71% of the 172 validated errors originated from two upstream system integration points—a figure computed on the validated set. This pattern was invisible in manual reports, which aggregated errors by type without preserving the source information needed to trace them upstream. Table 1 summarizes the comparison.

Table 1. Manual vs. AI-Assisted Reconciliation: Comparative Output (6-Week Cycle, ~84,000 Transactions)

Attribute	Manual Process	AI-Assisted (QuickSight)
Error detection method	Analyst review of line-item exports; discrepancies flagged individually per reporting cycle	Pattern recognition across full dataset (~84,000 transactions); automated anomaly clustering by source
Raw flags identified (6-week cycle)	147 discrepancies reported	203 discrepancies flagged — 38% more than manual count (raw, pre-validation)
Validated errors (post-reclassification)	147 (manual process not designed to isolate timing artifacts)	172 confirmed errors after 15% reclassification of timing artifacts — ~17% net increase over manual count
Root-cause visibility	Errors aggregated by type; upstream source not traced	71% of validated errors traced to two upstream integration points (computed on validated 172)

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Time to produce report	~3 analyst-days per cycle (based on logged analyst hours)	~4 hours analyst review after QuickSight processing (directional; not formally audited)
Advisor-facing output	Error count with type classification; no actionable root cause	Root-cause analysis with integration-point identification and remediation path

The compliance dimension of this finding warrants explicit acknowledgment. Discovering that prior reconciliation reports sent to financial advisors had systematically under-counted errors is not merely a data quality issue. In a regulated environment, such a discovery may carry disclosure or remediation obligations depending on the materiality of the under-count and the applicable regulatory framework. In this case, the team prepared a structured transition brief for advisors explaining the prior methodology and its limitations, and legal counsel was consulted on whether formal regulatory disclosure was required. Given the size and nature of the errors, no regulatory filing was triggered—but that determination was institution-specific and should not be assumed to generalize. Managers facing analogous discoveries should treat regulatory assessment as a mandatory early step, not an afterthought.

IV. What Went Wrong: Resistance and Its Roots

The deployment did not proceed smoothly. Within the first weeks, experienced analysts raised objections to QuickSight's findings. Some flags, they argued, reflected timing artifacts rather than genuine errors—a position that, as noted above, proved partially correct. However, the resistance extended well beyond those technically grounded objections.

This pattern is consistent with what Dietvorst et al. [4] documented experimentally: people abandon algorithms after observing even a single error more rapidly than they abandon human forecasters making identical mistakes. In this deployment, the 15% reclassification rate—a reasonable false-positive figure for an initial deployment—was sufficient to trigger skepticism that persisted beyond the specific flags in question. Logg et al. [5] further suggest that domain experts are particularly susceptible to this effect, as algorithmic advice threatens the professional identity built on the judgment the tool appears to supplant. That dynamic was visible here: the analysts most resistant to QuickSight were those with the deepest manual reconciliation experience.

Several analysts also reduced the frequency with which they updated the data fields feeding the QuickSight model. This observation is offered as a directional assessment rather than a measured causal claim; data quality degradation was noted by the implementation team but not formally quantified. The pattern is consistent with Leonardi's [6] finding that negative interpretations formed through early material interactions with a technology can stymie adoption even among those who endorse the organization's broader improvement goals.

The corrective that ultimately worked was reframing QuickSight's outputs as candidates for analyst review rather than automated verdicts. The 15% reclassification rate became evidence of analyst expertise adding genuine value—not evidence of the tool's unreliability. Once that framing took hold, engagement stabilized and data quality in the system recovered.

V. Managerial Implications

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The following implications are organized by organizational context. The case organization was a mid-sized registered investment advisory firm; the SME and startup guidance below represents reasoned extrapolation informed by the case findings and the regulatory landscape, not direct observation.

Large Financial Institutions. In large organizations, the primary challenge is not access to AI tooling but change management at scale. Manual reconciliation processes are often embedded in audit trails, regulatory submissions, and internal governance structures. Managers should plan for a calibration period of two to three reporting cycles before AI-assisted outputs are used for regulatory submission, and should document the transition methodology explicitly. Governance should specify which error flags require mandatory analyst review before action, preventing both over-reliance on AI output and accountability diffusion. Given SEC and FDIC requirements [1], [2], formal legal and compliance sign-off is advisable before live deployment. If AI-assisted analysis reveals that prior reports under-counted errors, regulatory counsel should assess disclosure obligations before any stakeholder communication.

Small- and Medium-Sized Enterprises. SMEs frequently lack dedicated data engineering teams, so AI-assisted reconciliation tools must be selected for configurability without deep technical customization. Cloud-based platforms such as QuickSight offer consumption-based pricing accessible without significant capital investment, but their value depends entirely on data pipeline integrity. The most common failure mode is connecting a BI tool to fragmented or inconsistently maintained data sources. Managers at SMEs should conduct a data audit before tool selection and consider a single-use-case entry strategy—one account type, one system integration pair—before broader rollout. Demonstrated success on the narrow case builds team confidence and establishes return on investment before scaling.

Fintech Startups. As the FDIC tightens recordkeeping requirements for bank-fintech partnerships [7], the compliance cost of manual reconciliation is rising. Startups that build AI-assisted reconciliation into their data infrastructure from inception avoid the organizational change management challenges of retrofitting established manual processes. The key risk is over-automation: deploying AI reconciliation without human review checkpoints creates audit trail gaps that regulators may challenge. A lightweight governance framework—even a two-person review loop on high-severity flags—is advisable from day one. Startups should also document their AI-assisted reconciliation methodology for investor and banking partner due diligence.

Across all three contexts, two prerequisites consistently determine whether AI-assisted reconciliation delivers on its potential: data pipeline integrity (the tool is only as reliable as the data it ingests) and analyst involvement in calibration (domain knowledge held by experienced analysts must be structured into the validation workflow, not bypassed by it). Success should be measured in validated error accuracy, analyst time reallocation to higher-value work, advisor confidence in reporting quality, and regulatory defensibility of the methodology.

VI. Conclusion

Moving from manual to AI-assisted data reconciliation in financial services is a management project in which technology plays a central role. The QuickSight deployment described in this article validated 172 errors compared with 147 from the manual process—a net increase of approximately 17%—and traced 71% of those validated errors to two upstream integration points the prior approach had never isolated. Those outcomes were meaningful. They were achieved only

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after addressing the algorithm aversion, organizational resistance, and governance gaps that the initial deployment had not anticipated.

The practical guidance offered here reflects the reality that the stakes, resources, and regulatory contexts of large institutions, SMEs, and startups differ materially. What does not differ is the leadership challenge: creating conditions in which an AI tool's ability to see more, and to see faster, translates into better decisions for the clients and stakeholders who depend on accurate financial data.

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Managerial Implications Statement

This article provides engineering and technology managers in financial services with a practitioner-grounded framework for deploying AI-assisted reconciliation tools. Implications are differentiated for large financial institutions (change management at scale, regulatory sign-off, compliance disclosure obligations on discovering under-counted errors), small- and medium-sized enterprises (data pipeline prerequisites, single-use-case entry strategy, consumption-based tooling), and fintech startups (FDIC compliance drivers, lightweight governance from day one, investor due diligence documentation). Success metrics—validated error accuracy, analyst time reallocation, advisor confidence, and regulatory defensibility—are identified to support ongoing investment evaluation.

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Biography

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